

Abdullah Tahir

AI/ML Engineer

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in AI/ML, with hands-on experience independently building and shipping AI applications — RAG pipelines, agentic workflows, and LLM-powered tools — using Python, LangChain, and LangGraph. Comfortable across the full AI project stack: prompt design, vector search, backend API design, and production-style data modeling. Seeking an AI/ML Engineer internship to bring these skills to a real engineering team and grow under experienced mentors.

TECHNICAL SKILLS

- **AI / ML:** LangChain, LangGraph, LangSmith, RAG, LLM Fine-tuning, Prompt Engineering, Agentic AI, Hugging Face, Ollama, Qwen2.5
- **Languages:** Python, OOP
- **Data & Libraries:** Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib
- **Databases & Search:** PostgreSQL, SQLAlchemy, Alembic, Vector Databases (FAISS / Chroma / pgvector), Embeddings, Semantic Search
- **Web & APIs:** FastAPI, Streamlit, Pydantic, JWT Auth, REST API Integration (Requests), OpenRouter, PyPDF
- **Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook, Docker (basic)

PROJECTS

Research Intelligence Platform

2026 – Present

Python, FastAPI, PostgreSQL, pgvector, LangChain, LangGraph, JWT | [GitHub](#)

- Architected a full-stack AI research platform for discovering, managing, and understanding research papers, with secure document management including JWT authentication, ownership protection, search, filtering, sorting, pagination, and DOI uniqueness enforcement
- Built an end-to-end RAG pipeline spanning PDF processing, text chunking, and 384-dimensional embeddings, using PostgreSQL/pgvector for semantic search and similarity-based retrieval to ground LLM-generated responses in source content
- Integrated LangGraph agent workflows with tool calling, enabling the agent to autonomously decide between retrieving information from a paper and responding directly to general queries
- Implemented persistent, thread-based conversation memory via PostgreSQL checkpointing, and designed the architecture to extend toward future agentic research capabilities such as paper discovery, analysis, summarization, and recommendations

AI Career Mentor

2026

Python, Streamlit, Google Gemini API, Google GenAI SDK, python-dotenv | [GitHub](#)

- Built an AI-powered career guidance chatbot for Computer Science students and early-career developers, delivering personalized career planning, learning roadmaps, AI/ML project recommendations, and internship and technical interview preparation
- Designed a custom system prompt to establish a dedicated career-mentor persona, keeping the Gemini-powered assistant's responses focused and relevant to technology and career development
- Implemented an interactive Streamlit chat interface with persistent in-session conversation history, clear-chat functionality, and robust error handling for API failures
- Secured API credentials using environment variables via python-dotenv, following secure configuration practices suitable for production-style deployment

AI Data Analyst Assistant

2026 – Present

Python, FastAPI, Streamlit, Pandas, Seaborn, Pydantic, Ollama | [GitHub](#)

- Built a full-stack AI data analysis platform with a Streamlit frontend and FastAPI backend that ingests CSV datasets of 10,000+ rows, generating statistical summaries, correlation heatmaps, and LLM-generated business insights
- Implemented a natural-language Q&A interface over uploaded datasets with sliding-window context management for multi-turn queries; used Pydantic for typed request/response validation across the FastAPI layer
- Designed a modular, production-style project architecture separating data processing, API layer, and AI inference — enabling clean component boundaries and maintainable code at scale

Agentic AI Chatbot — LangGraph + RAG

2025

Python, LangGraph, LangChain, LangSmith, FAISS, Hugging Face, OpenRouter, Streamlit | [GitHub](#)

- Built an agentic chatbot using LangGraph that routes between direct LLM generation and RAG-based PDF retrieval via tool calling, supporting documents 100+ pages and maintaining context across 20+ conversation turns
- Implemented a chunking strategy (500-token chunks, 50-token overlap) with a FAISS vector store and Hugging Face embeddings, achieving retrieval latency under 2 seconds and grounding answers in retrieved chunks to reduce hallucination
- Implemented persistent conversation memory via LangGraph Checkpointer for session resumption; integrated LangSmith for end-to-end agent tracing and connected OpenRouter API for flexible LLM backend switching without code changes

EDUCATION

Air University, Islamabad, Pakistan

2024 – Present

Bachelor of Science in Computer Science (BSCS) — Specialization: Artificial Intelligence & Machine Learning

- Relevant coursework: Python Programming, Object-Oriented Programming, Introductory AI Concepts, Data Structures

CERTIFICATIONS & CONTINUING EDUCATION

- Deeplearning.ai — Prompt Engineering for Developers
- Google — AI Essentials
- Active learning: LLM fine-tuning, Hugging Face Transformers, Agentic AI systems, Docker containerization for ML apps